

Research on Aerial Autonomous Docking and Landing Technology of Dual Multi-rotor UAV

Liang Wang✉, Xiangqian Jiang, Di Wang, Lisheng Wang and Zhijun Tu
Shanghai Aerospace Systems Engineering Institute, Shanghai and 201108, China

✉Email: ckckli985@gmail.com

This study conducted an in-depth investigation of the tracking controller of an aerial autonomous docking and landing trajectory system. To examine the system’s overall design, a dual-machine trajectory-tracking control simulation platform was created via MATLAB Simulink. Then, an autonomous docking and landing trajectory-tracking controller based on backpropagation proportional–integral–derivative control was designed, which fulfilled the trajectory-tracking control requirements in the autonomous docking and landing process by efficiently suppressing the external airflow disturbance according to the simulation results. A YOLOv3-based vision pilot system was designed to calibrate the rate of the aerial docking and landing position to eight frame per second. The feasibility of the multi-rotor aerial autonomous docking and landing system was verified using prototype flight tests during the day and at night.

Introduction: The range, hang time and types of tasks performed by small-rotor unmanned aerial vehicles (UAVs) are greatly limited by the energy densities of lithium batteries, which are often used to drive small multi-rotor UAVs[1-4]. To overcome the above technical limitations, a cluster technical scheme was adopted to design a platform UAV that can carry a Mission UAV. A small quadrotor was used as the Mission UAV, and a large hexacopter UAV was used as the Mission UAV take-off and landing platform; the hexacopter was also the Mission UAV carrier. Although difficult, aerial autonomous docking and landing were accomplished.

Modelling of Mission UAV and UAV Platform: By analysing the mission requirements in detail, Mission UAV and the UAV platform were designed. First, the mathematical models were derived and established by analysing the frame design, airborne equipment type selection and power system matching. Based on this analysis, an experimental prototype was fabricated to verify the aerial autonomous docking, take-off and landing technology, as shown in Fig. 1 to Fig. 4. Fig. 5 shows the designed flight simulation environment.

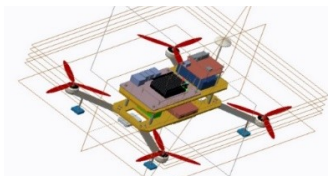


Fig. 1 Three-dimensional model of Mission UAV

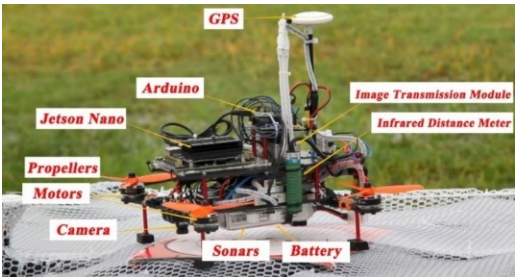


Fig. 2 Real Mission UAV

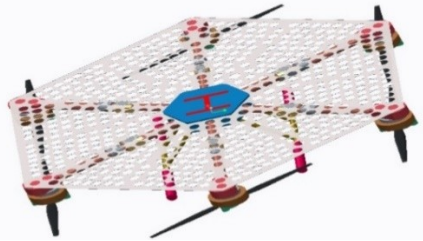


Fig. 3 Three-dimensional model of the UAV platform



Fig. 4 Real UAV platform

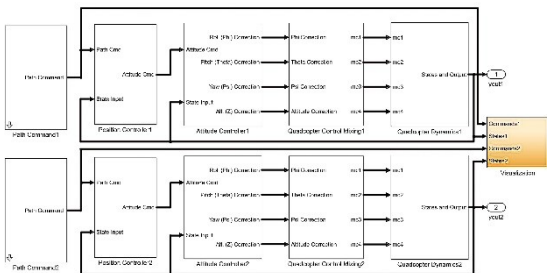


Fig. 5 Flight simulation environment

Architectural Design of Aerial Autonomous Docking and Landing Control System: Fig. 5 shows the architectural design of the aerial autonomous docking and landing control system. First, the video information stream was sent to the image processing module by image acquisition to identify, locate and track the landing area of the UAV platform. The data obtained from positioning after calculating the vertical and horizontal displacements were sent to the single-chip microcomputer module to generate the expected trajectory control signal of the vertical and horizontal displacements of Mission UAV. Next, the control signal obtained was input to the obstacle avoidance system for safeguarding the UAV flight environment; the signal was also input to the underlying flight control system to control Mission UAV for the expected trajectory-tracking movement. The visual feedback was conducive to quickly updating the location of the UAV platform and updating the rate in real time, which was the loop execution rate for the autonomous docking and landing control system.

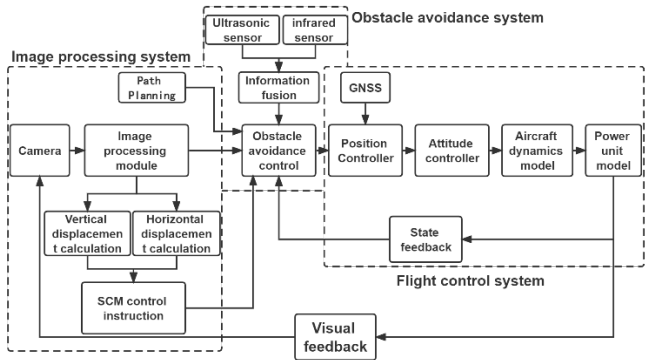


Fig. 5 Architectural design for aerial autonomous docking and landing control system

Eight Frames per Second Trajectory-Tracking Controller: Fig. 6 shows the proportional–integral–derivative (PID) controller structure based on the backpropagation (BP) network[5]; it has two main parts: a classic

controller and a neural network. The classic controller can achieve the expected closed-loop control effect with the three data elements k_p , k_i and k_d as its online control modes. The BP neural network can adjust data by referring to the actual operational status[6]. In this way, the network can achieve the expected performance improvement; that is, the actual output status can correspond to the three adjustable data elements k_p , k_i and k_d of PID. Moreover, the matched control measures that focus on weighting the coefficients were used to obtain the ideal control data of PID for the actual steady condition according to the deep learning of the neural network.

The simulation time was set to 160 s, and the fixed step size $t=0.02$ s was the solver sampling period. The initial values of the K_{p0} , K_{i0} and K_{d0} parameters were $K_{p0}=0.6$, $K_{i0}=0.01$ and $K_{d0}=0.2$. Fig. 7 shows the simulation result.

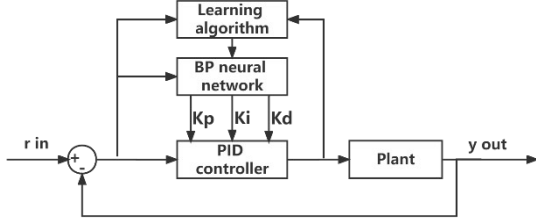


Fig. 6 Adaptive PID control system based on the BP neural network

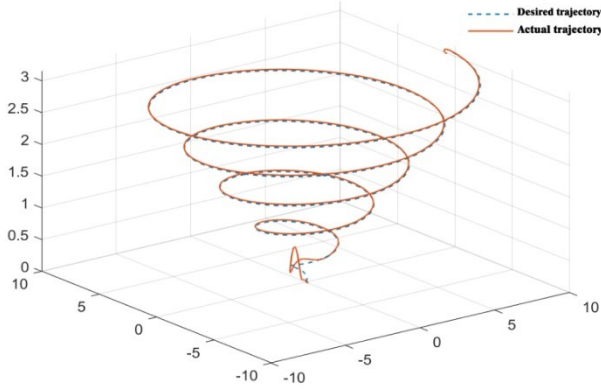


Fig. 7 Three-dimensional trajectory tracking based on BP-PID control

Velocities, angles and position response curves of the controller based on BP-PID were simulated in various directions, as shown in Fig. 7. The actual flight trajectory with high accuracy in tracking the expected trajectory can be observed from the position response curves in the X, Y and Z directions. Mission UAVs based on BP-PID control can realise high-precision 3D trajectory-tracking performance, as shown in the Fig.8 .

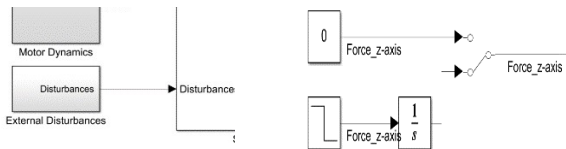


Fig. 8 Continuous cumulative disturbance applied along the Z-axis of the geographic coordinate system of Mission UAV in the first second

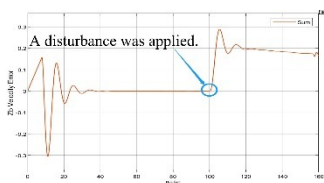


Fig. 9. Variation curve of velocity error along the Z-axis under the

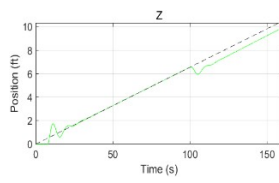


Fig. 10 Expected trajectory and actual trajectory along

applied disturbance

the Z-axis under the applied disturbance

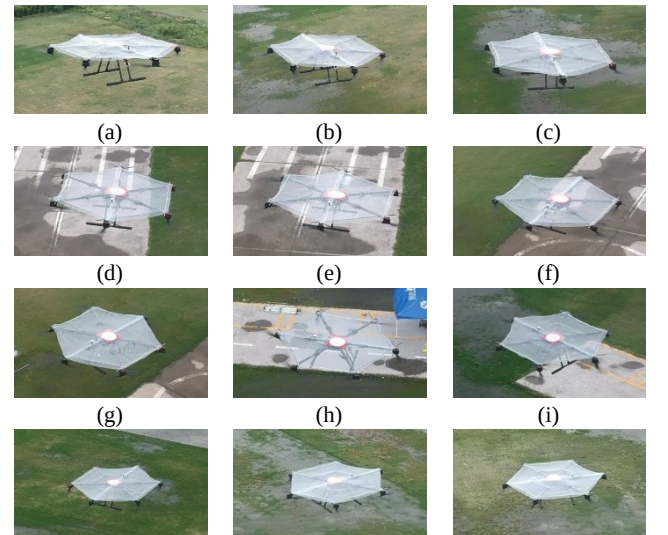
The step signal module and the integral module were introduced into the simulation platform to simulate the disturbance of the blade airflow on the UAV platform because of the continuously downward increasing force of the Mission UAV landing on the UAV platform. A disturbance was applied to the design after the 100th second of the simulation cycle. Figs. 9 and 10 show that the variations of the velocity error increased significantly in the Z direction. The error range was kept at 0.3 by using the BP-PID controller. Then, the error value was suppressed continuously, and it diminished gradually. Fig. 11 shows that a certain error value was generated between the position tracking in the Z direction and the expected trajectory in the 100th second after a continuously increasing disturbance was applied downward; the error value was under control. Evidently, the BP-PID controller has practical significance in enhancing the trajectory-tracking performance and the robustness of the control system of Mission UAV.

Design of Vision Pilot System Based on YOLOv3: The main idea of the you-only-look-once (YOLO) system is to solve the target detection problem as a regression problem. YOLOv3 accesses the network by directly inputting the entire image and then directly regressing the frame position and the target category[7, 8]. The YOLOv3 network divides the input image into $S \times S$ grid cells. If the geometric centre of a target falls in a cell, the cell will be responsible for detecting the target. Each cell takes charge of predicting B bounding boxes and the confidence. Here, B is the number of categories identified by the target in the dataset, and the confidence reflects the regression accuracy of the predicted position of the bounding box. The accuracy is the product of the probability of the target being detected and the intersection ratio of the bounding box and the real position (see Equation (1)). The confidence is defined as follows:

$$\text{Confidence} = P_r(\text{Object}) \times \text{IoU}_{\text{truth pred}} \quad (1)$$

Here, $\text{Pr}(\text{Object})$ is the possibility of the target existing in the box. If there is no object in the box, then $\text{Pr}(\text{Object})=0$; if there is an object, then $\text{Pr}(\text{Object})=1$.

The YOLOv3 network that features a strong capability of feature expression and transfer can learn to detect an object in the detection image upon training with large amounts of image training sets that contain a certain object. We expected that YOLOv3 would detect only the target area on the UAV platform in the image detection of the video image. Therefore, the network was trained on the constructed dataset of the target area on the prepared platform where the UAV landed. Then, the trained network could detect and identify the target area in the video stream. Fig. 11 shows the training set of positive samples in the landing area.



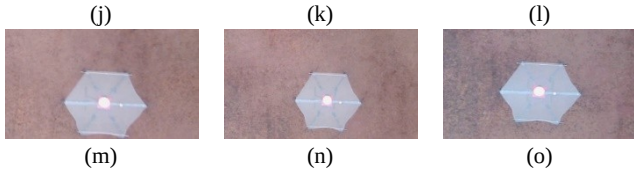


Fig. 11 Training set of positive samples in the landing area

Aerial Autonomous Docking and Landing Flight Experiment: The designed aerial autonomous docking and landing system operated well in the field flight test. The docking and landing system recorded average system identification, positioning and tracking during the day, which reached 8.83 FPS and 8.24 FPS, respectively, at night. The distribution of the landing points of Mission UAV on the UAV platform reached 15 cm on average in comparison with the distance accuracy from the centre point of the landing surface of the UAV platform. Clearly, design expectations were satisfied in the field test. Fig. 12 shows the aerial autonomous docking and landing test image in the field (see the video screenshot), and Fig.13 shows the aerial autonomous docking and landing test image under the infrared mode at night (see video snapshot of the onboard image processing system of the Mission UAV).

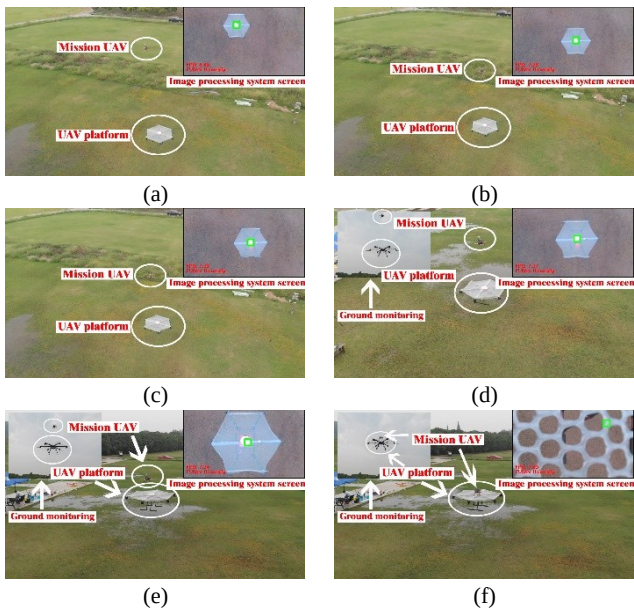


Fig. 12 Image of aerial autonomous docking and landing test (video snapshot)

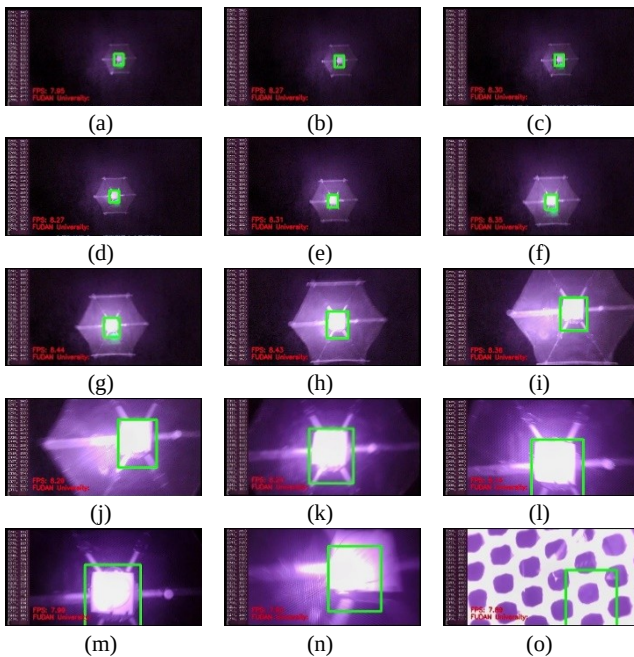


Fig. 13 Aerial autonomous docking and landing test under infrared

mode at night (video snapshot of the onboard image processing system of the Mission UAV)

Conclusion: This study examines the aerial autonomous docking take-off and landing technology based on multiple rotors. The precise autonomous docking and landing process control of the Mission UAV was implemented on the UAV platform. First, a simulation platform was built in the MATLAB Simulink environment; the design was that of a preset expected trajectory according to the overall design and modelling results of Mission UAV and the UAV platform. Then, an autonomous docking and landing trajectory tracker based on BP-PID control was designed with the flight control influence of the rotor airflow disturbance of the UAV platform used as a key interference factor in the dual-UAV autonomous docking and landing process. The influence of airflow disturbance on the control performance of the Mission UAV was simulated by the introduction of a continuously changing disturbing quantity. The BP-PID control technology conformed with the tracking accuracy requirements of the landing trajectory under the influence of the disturbance. Moreover, a visual pilot system based on YOLOv3 was also designed to ensure that the real-time image positioning calibration rate could reach 8 FPS during the dynamic docking and landing process of Mission UAV on the UAV platform. This provides the real-time positioning requirements of the landing area where Mission UAV landed in the dynamic take-off and landing process. Finally, flight experiments were conducted. The control stability and positioning accuracy of take-off and landing were proven to conform with the design expectations based on the dynamic take-off and landing tests conducted during the day and at night under the infrared mode.

Flight experiment video: <https://youtu.be/rCrgNV2GRFg>

Conflict of interest: The authors declare no conflict of interest.

Data availability statement: This study is the contribution of author Liang Wang during his doctoral study in Fudan University, and is now planned to be compiled and published. All the original experimental data and original experimental videos of this study can be obtained by contacting the corresponding author Liang Wang. Researchers are welcome to conduct academic exchanges.

References

1. Cesare, K., Skeeel, R., Yoo, S.-H., Zhang, Y., and Hollinger, G., 'Multi-Uav Exploration with Limited Communication and Battery', in, 2015 IEEE international conference on robotics and automation (ICRA), (IEEE, 2015)
2. Dudek, M., Tomczyk, P., Wygonik, P., Korkosz, M., Bogusz, P., and Lis, B.J.I.J.E.S., 'Hybrid Fuel Cell-Battery System as a Main Power Unit for Small Unmanned Aerial Vehicles (Uav)', 2013, 8, (6), pp. 8442-8463.
3. Abeywickrama, H.V., Jayawickrama, B.A., He, Y., and Dutkiewicz, E.J.I.a., 'Comprehensive Energy Consumption Model for Unmanned Aerial Vehicles, Based on Empirical Studies of Battery Performance', 2018, 6, pp. 58383-58394.
4. Ahani, G., Yuan, D., and Zhao, Y.J.I.C.L., 'Age-Optimal Uav Scheduling for Data Collection with Battery Recharging', 2020, 25, (4), pp. 1254-1258.
5. Patel, R. and Kumar, V.J.P.C.S., 'Multilayer Neuro Pid Controller Based on Back Propagation Algorithm', 2015, 54, pp. 207-214.
6. Son, N.N., Van Kien, C., Anh, H.P.H.J.R., and Systems, A., 'A Novel Adaptive Feed-Forward-Pid Controller of a Scara Parallel Robot Using Pneumatic Artificial Muscle Actuator Based on Neural Network and Modified Differential Evolution Algorithm', 2017, 96, pp. 65-80.
7. Redmon, J. and Farhadi, A.J.a.p.a., 'Yolov3: An Incremental Improvement', 2018.
8. Zhao, K., Zhu, X., Jiang, H., Zhang, C., Wang, Z., and Fu, B.J.E.L., 'Dynamic Loss for One-Stage Object Detectors in Computer Vision', 2018, 54, (25), pp. 1433-1434.